

2024 Ph H2 Q1

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

Summary of Question: A 'Doppler ball' is thrown from student A at speed 11.0 m s^{-1} at 36.0° from a height of 1.60 m ; it passes over student B and lands at point Q. (a) (i) Find horizontal and vertical components of the initial velocity. (ii) Given the time to Q is 1.53 s , find the horizontal distance. (iii) At $t = 0.95 \text{ s}$ the ball is above student B who can reach 2.10 m ; determine the height gap h . (b) The ball later moves at 8.60 m s^{-1} towards B along ropes while emitting 622 Hz ; (i) find the frequency heard by B, (ii) explain how the foam protects the circuit board.

(a)(i) Components of initial velocity

Relationship: $u_x = u \cos\theta$, $u_y = u \sin\theta$

Substitution: $u_x = 11.0 \times \cos 36.0^\circ = 8.90 \text{ m s}^{-1}$; $u_y = 11.0 \times \sin 36.0^\circ = 6.47 \text{ m s}^{-1}$

Final answers: $u_x = 8.90 \text{ m s}^{-1}$, $u_y = 6.47 \text{ m s}^{-1}$

(a)(ii) Horizontal distance to Q

Relationship: $s = v t$ (horizontal motion at constant speed u_x)

Substitution: $s = 8.90 \times 1.53 = 13.62 \text{ m}$

Final answer: 13.6 m (to 3 s.f.)

(a)(iii) Height above student B at $t = 0.95$ s

Relationship: $y(t) = y_0 + u_y t - \frac{1}{2} g t^2$; $h = y(t) - 2.10$

Substitution: $y(t) = 1.60 + 6.47 \times 0.95 - 0.5 \times 9.8 \times (0.95)^2 = 3.320$ m; $h = 3.320 - 2.10 = 1.220$ m

Final answer: $h = 1.22$ m (to 3 s.f.)

(b)(i) Doppler frequency heard by B

Relationship (source approaching stationary observer): $f' = f \frac{v_{\text{sound}}}{(v_{\text{sound}} - v_{\text{source}})}$

Substitution: $f' = 622 \times 340 / (340 - 8.60) = 638.1$ Hz

Final answer: 638 Hz (to 3 s.f.)

(b)(ii) Why the foam protects the circuit board

During the collision the foam increases the contact time, so for the same change of momentum (Δp), the average force $F = \Delta p / \Delta t$ is smaller. Hence a smaller force acts on the circuit board.

Revision Tips

- Components: $u_x = u \cos\theta$, $u_y = u \sin\theta$. Horizontal motion is constant speed; vertical motion has constant acceleration (g).
- For projectiles, keep a clear sign convention: take upward as +ve so $a = -g$.
- Doppler (source $\rightarrow$ observer): $f' = f v/(v - v_s)$. Increasing contact time reduces impact force: $F = \Delta p/\Delta t$.